

CA - Experiment 5 - Introduction to Jenkins: What is Jenkins?, Installing Jenkins on Local or Cloud Environment, Configuring Jenkins for First Use

Experiment 5: Introduction to Jenkins

🌟 Objective

To understand the fundamentals of **Jenkins**, install it on a local or cloud environment, and configure it for first-time use.

♦ 1. What is Jenkins?

Jenkins is an **open-source automation server** used for:

- ✓ **Continuous Integration (CI)** – Automatically testing and integrating code changes
- ✓ **Continuous Deployment (CD)** – Automating application deployment
- ✓ **Building Pipelines** – Managing end-to-end software development workflows
- ✓ **Plugin-Based Extensibility** – Supporting tools like Maven, Gradle, Ansible, Docker, and Azure DevOps

🚀 Why Use Jenkins?

- ✓ Automates builds and tests
- ✓ Reduces manual intervention
- ✓ Improves software quality
- ✓ Works with multiple tools and platforms

♦ 2. Installing Jenkins

Jenkins can be installed using multiple methods:

- 1 **Windows Installer (.msi) – Recommended for Windows**
- 2 **Linux Package Manager – Best for Linux Users**
- 3 **Jenkins WAR File – Universal method using Java**

We'll cover all three approaches.

● 2.1 Prerequisites

- ♦ **Java 11 or 17** is required for Jenkins.

Check Java version:

```
1 java -version
2
```

If Java is not installed, download it from:

[Download the Latest Java LTS Free](#)

2.2 Installing Jenkins on Windows (MSI Installer) – Recommended

✓ Step 1: Download Jenkins

🔗 Download from: [Download and deploy](#)

Choose **Windows Installer (.msi)** for an easy setup.

✓ Step 2: Install Jenkins

- 1 Run the downloaded `.msi` file.
- 2 Follow the installation wizard.
- 3 Select **Run Jenkins as a Windows Service** (recommended).
- 4 Choose the **installation directory** (default: `C:\Program Files\Jenkins`).
- 5 Click **Install** and wait for the setup to complete.

✓ Step 3: Start Jenkins

- 1 Open **Services** (`services.msc`) and ensure **Jenkins** is running.
- 2 Open a web browser and go to:

```
1 http://localhost:8080
2
```

✓ Step 4: Unlock Jenkins

- 1 Find the initial **Admin Password** in:

```
1 C:\Program Files\Jenkins\secrets\initialAdminPassword
2
```

- 2 Copy the password and paste it into the Jenkins setup page.

✓ Step 5: Install Recommended Plugins

Jenkins will prompt you to install plugins. Click **"Install Suggested Plugins"**.

✓ Step 6: Create Admin User

- 1 Set up a **Username**, **Password**, and **Email**.
- 2 Click **Save and Finish**.

♦ **Jenkins is now ready!** 🎉

Access it anytime at:

```
1 http://localhost:8080
2
```

2.3 Installing Jenkins on Linux (Ubuntu/Debian)

✓ Step 1: Add Jenkins Repository

```
1 wget -q -O - https://pkg.jenkins.io/debian/jenkins.io.key | sudo apt-key add -
2 sudo sh -c 'echo deb http://pkg.jenkins.io/debian-stable binary/ >
   /etc/apt/sources.list.d/jenkins.list'
3
```

✓ Step 2: Install Jenkins

```
1 sudo apt update
```

```
2 sudo apt install jenkins -y
3
```

✓ Step 3: Start Jenkins

```
1 sudo systemctl start jenkins
2 sudo systemctl enable jenkins
3
```

✓ Step 4: Access Jenkins

Find the **initial password** in:

```
1 sudo cat /var/lib/jenkins/secrets/initialAdminPassword
2
```

Then open Jenkins in a browser:

```
1 http://localhost:8080
2
```

🟡 2.4 Installing Jenkins Using WAR File (Works on Any OS)

This method allows you to run Jenkins without installing it as a service.

✓ Step 1: Download the Jenkins WAR File

🔗 Download from: [Download and deploy](#)

Choose **Generic Java Package (.war)**.

✓ Step 2: Run Jenkins Using Java

Navigate to the folder where the `.war` file is downloaded and run:

```
1 java -jar jenkins.war --httpPort=8080
2
```

♦ This will start Jenkins on port **8080**.

✓ Step 3: Open Jenkins in Browser

Go to:

```
1 http://localhost:8080
2
```

✓ Step 4: Unlock Jenkins & Setup

Follow the **same steps** as the Windows/Linux installation:

- ✓ Find the initial password
- ✓ Install plugins
- ✓ Create an admin user

♦ **Jenkins is now running without installation!**

To stop Jenkins, press **CTRL + C** in the terminal.

♦ 3. Configuring Jenkins for First Use

✓ 3.1 Understanding the Jenkins Dashboard

After logging in, you will see:

- ♦ **New Item** → Create Jobs/Pipelines
- ♦ **Manage Jenkins** → Configure System, Users, and Plugins
- ♦ **Build History** → View previous builds
- ♦ **Credentials** → Store secure authentication details

✓ 3.2 Installing Additional Plugins

Jenkins supports **plugins** for various tools like Maven, Gradle, Docker, and Azure DevOps.

- ♦ To install a plugin:

- 1 Go to **Manage Jenkins** → **Manage Plugins**
- 2 Search for the required plugin
- 3 Click **Install without Restart**

✓ 3.3 Setting Up Global Tool Configuration

Configure Java, Maven, and Gradle in Jenkins:

- 1 Go to **Manage Jenkins** → **Global Tool Configuration**
- 2 Add paths for:
 - **JDK** (C:\Program Files\Java\jdk-17)
 - **Maven** (C:\Maven\apache-maven-<version>)
 - **Gradle** (C:\Gradle\gradle-<version>)
- 3 Click **Save**

📌 Assessment Questions

- 1 What is **Jenkins** used for in DevOps?
- 2 Explain the **difference** between CI and CD in Jenkins.
- 3 How do you **install Jenkins** on Windows, Linux, and using the WAR file?
- 4 Where can you find the **initial Jenkins password** after installation?
- 5 What are some **essential Jenkins plugins** for automation?

📌 Important Note

You can use the below links to easily install **Jenkins** & understand **CI/CD Pipelines** in **DevOps**.

- ♦ **Jenkins Installation Video** 📺 – Follow this step-by-step guide for a **seamless Jenkins setup**: [Jenkins Installation - Step by Step Guide](#)

- ♦ **Understanding CI/CD in DevOps** 🚀 – Learn how Jenkins fits into the **CI/CD pipeline**: [DevOps CI CD Pipeline | Simple & Detailed Explanation](#)

Ensure you have **Java installed** before setting up Jenkins. If you face any issues, check the **Jenkins logs** for troubleshooting. ✓